

# ACTION CHUNK SEAM BOUNDARY: DERIVATIVE CONTINUITY VS. TRANSMISSION SHOCK

Systems Dynamics of Unchecked  $C^0$  Concatenation vs. Jerk-Continuous  $C^2$  Quintic Spline Trajectory Blending

## ACTIVE ACTION CHUNK A

Horizon:  $t \in [0, 50 \text{ ms}]$  (20 Hz)  
 $q_A(50) = 0.850 \text{ rad}$   
 $\dot{q}_A(50) = -0.45 \text{ rad/s}$  (Decel)

## VLA POLICY INFERENCE (BRAIN)

Request Lead:  $T_{\text{req}} = 0 \text{ ms}$   
Tail Latency:  $L \sim P99 = 48 \text{ ms}$   
Chunk B arrives at  $t = 48 \text{ ms}$

## CASE A: Naive $C^0$ Concatenation

Pos matched;  $\Delta \dot{q} = +0.60 \text{ rad/s}$  step!

## CASE B: $C^2$ Jerk-Continuous Spline

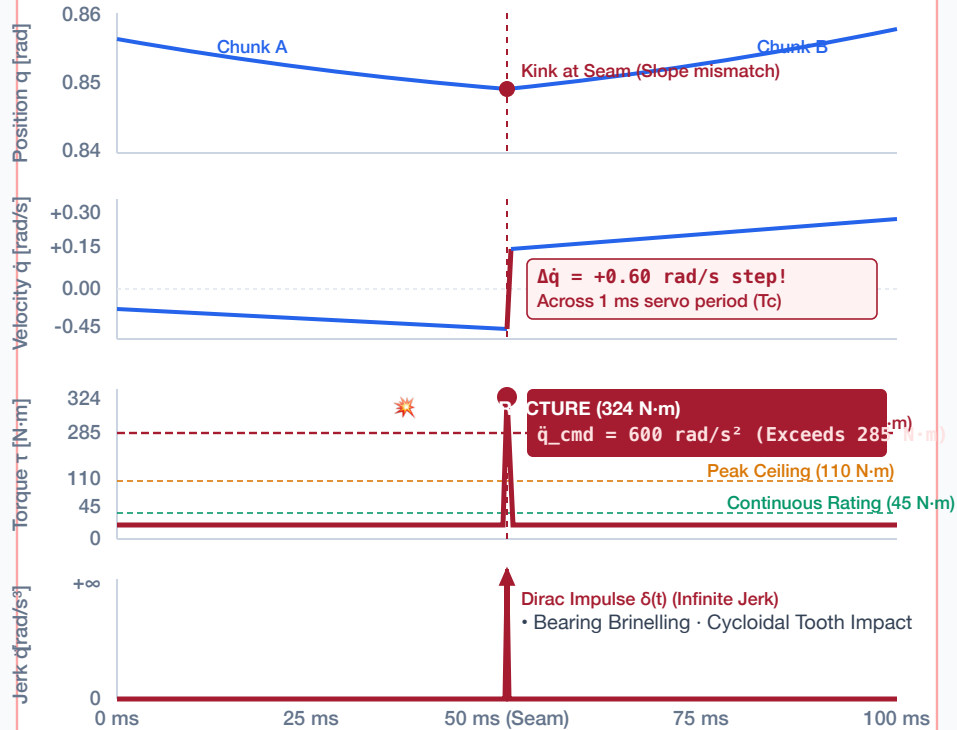
Quintic bridge over  $T_{\text{blend}} = 20 \text{ ms}$

## 1 kHz SERVO LOOP & PLANT

Servo Period:  $T_c = 1.0 \text{ ms}$   
Reflected Inertia:  $J = 2.7 \text{ kg} \cdot \text{m}^2$   
Pin Shear Limit:  $\tau = 285 \text{ N} \cdot \text{m}$

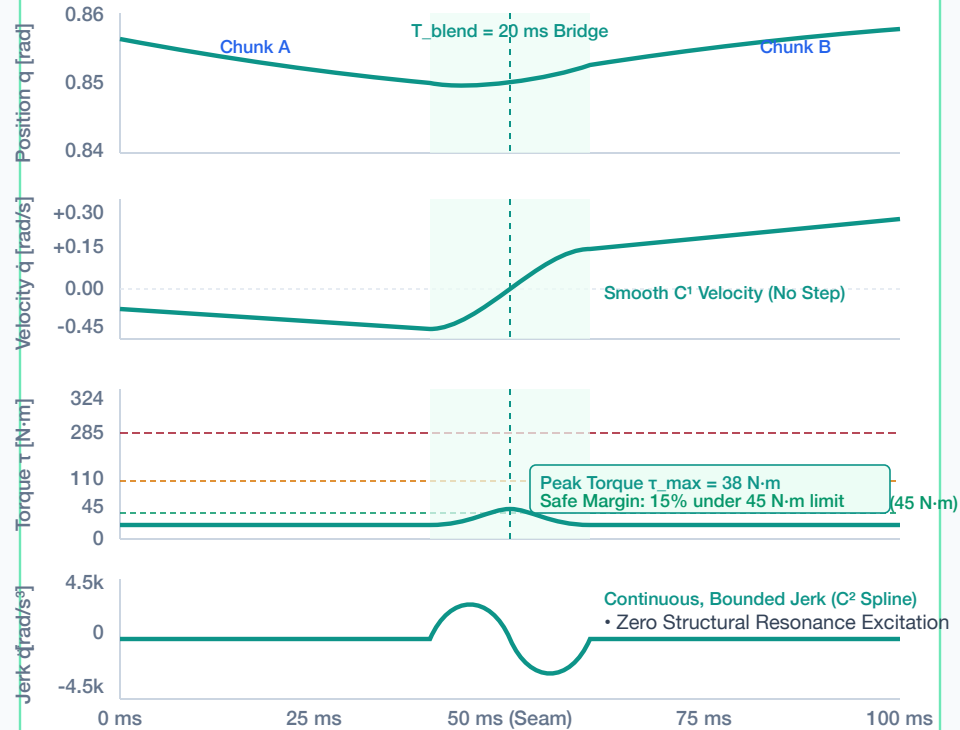
### CASE A: UNCHECKED $C^0$ POSITION-ONLY CONCATENATION

$C^0$  FRACTURE



### CASE B: JERK-CONTINUOUS $C^2$ QUINTIC BLENDED SPLINE

$C^2$  VERIFIED



## SYSTEMS IMPLICATION: CONCATENATION REQUIRES HIGHER-ORDER BOUNDARY VALIDATION

When an action chunk finishes, splicing a replacement that matches only position ( $C^0$ ) creates an instantaneous velocity step  $\Delta \dot{q}$ .

Across a 1.0 ms servo cycle ( $T_c$ ), this step demands an acceleration impulse  $\ddot{q} = \Delta \dot{q} / T_c = 600 \text{ rad/s}^2$ , transmitting 324 N·m through reflected inertia ( $J = 2.7 \text{ kg} \cdot \text{m}^2$ ) and shearing the drive pin.